



Bachelor Level / First Year/ First Semester/ Science
Computer Science and Information Technology (CSc. 111)
(Digital Logic)
(NEW COURSE)

Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt any two questions:

(2×10=20)

1. ✓ Design a combinational circuit that generates 9's complement of a BCD number. (10)

2. Implement the following functions using PLA (10)

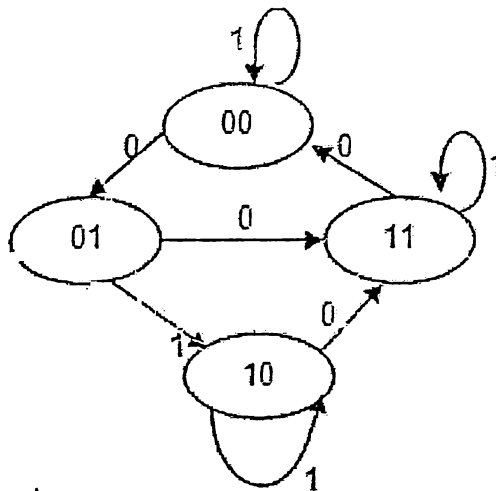
$$w(A, B, C, D) = \sum(2, 12, 13)$$

$$x(A, B, C, D) = \sum(7, 8, 9, 10, 11, 12, 13, 14, 15)$$

$$y(A, B, C, D) = \sum(0, 2, 3, 4, 5, 6, 7, 8, 10, 11, 15)$$

$$z(A, B, C, D) = \sum(1, 2, 8, 12, 13)$$

3. Design sequential circuit specified by the following state diagram using T flip-flops. (10)



Attempt any eight.

4. ✓ List two major characteristics of digital computer. Represent -6 (negative six) using 8 bits in signed magnitude, signed-1's-complement and signed-2's-complement respectively. Represent decimal number 4673 in a) octal, and b) BCD. (1+2+2)

5. ✓ Where is CMOS suitable to use? Define Power dissipation. Show that the positive logic NAND gate is a negative logic NOR gate and vice versa. (1+1+3)

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7. ✓ Design a full subtractor circuit with three inputs x , y , B_{in} and two outputs Diff and B_{out} . The circuit subtracts $x - y - B_{in}$ where B_{in} is the input borrow, B_{out} is the output borrow, and Diff is the difference. (5)
8. ✓ Design 4-bit even parity generator. (5)
9. ✓ What is the difference between a serial and parallel transfer? Explain how to convert serial data to parallel and parallel data to serial. What type of register is needed? (1+3+1)
10. Explain negative-edge triggered D flip flop with necessary logic diagram and truth table. (5)
11. ✓ Illustrate the use of Binary ripple counter and BCD ripple counter. (2.5+2.5)
12. ✓ Write Short notes on (Any two) (2x2.5)
- a) RTL
 - b) State Reduction
 - c) POS